

MODULE 2.4

Network Hardening: Segmentation

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Prepared by: Kudzaishe Majeza

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- 2.4.1 • What is Network Segmentation?
- 2.4.2 • Virtual Local Area Networks (VLANs)
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- 2.4.3 • Zones of Risk and the Zero Trust Model
- 2.4.4 • Managing Threats to the LAN

2.4.1 What is Network Segmentation?

Dividing a network into smaller parts for better security and performance

Network Segmentation Explained

- Segmentation involves dividing a computer network into smaller, isolated parts.
- Each segment acts as its own contained environment — traffic stays within the segment unless explicitly allowed to cross into another.
- This improves both network performance (less congestion) and security (an attacker in one segment cannot freely access other segments).
- For @Apollo, segmentation is a key strategy for protecting sensitive data and limiting the damage an attacker can do if they get in.

Why It Matters

- Without segmentation, every device on a network can potentially communicate with every other device.
- This means one compromised device can be used to attack everything else on the network.
- Segmentation contains the threat — isolating damage and making it far harder for an attacker to move laterally across the network.
- Common segmentation tools include VLANs, DMZs and firewalls.

2.4.2 Virtual Local Area Networks (VLANs)

Creating logical network segments to isolate and protect sensitive data

What is a VLAN?

- A VLAN is a virtual network created within a physical network using software configuration.
- Devices on a VLAN communicate as if they are on their own separate network, even if they share the same physical switches and cables.
- For @Apollo, VLANs allow the HR department's sensitive personnel data to be isolated from the rest of the network — even on the same building floor.

How VLANs Work & Security Considerations

- A network administrator assigns devices to VLANs based on department, floor, function or sensitivity level.
- Traffic between VLANs must pass through a router or firewall — it cannot flow freely between segments.
- VLANs also limit broadcast traffic, which improves overall network performance.
- Cybercriminals can attack VLANs through techniques like VLAN hopping, where they trick switches into forwarding traffic across VLAN boundaries.
- Countermeasures: Monitor VLAN performance, use advanced switch configurations, and regularly apply patches and updates to network equipment.

2.4.3 The Demilitarized Zone (DMZ)

A buffer network between the trusted internal network and the internet

What is a DMZ?

- A DMZ is a small, isolated network that sits between a trusted private network and the internet.
- It acts as a buffer zone — providing a controlled area where public-facing services can operate without exposing the internal network.
- Web servers and mail servers are typically placed in the DMZ so that external users can access them without ever reaching the internal network.

How the DMZ Protects the Network

- If an attacker compromises a server in the DMZ, they are still separated from the internal LAN by a firewall — containing the damage.
- Two firewalls are typically used: one between the internet and the DMZ, and one between the DMZ and the internal network.
- This two-firewall setup means an attacker must breach TWO security layers to reach sensitive internal systems.
- Common services hosted in the DMZ: web servers, email servers, FTP servers, DNS servers.
- Internal services that should NEVER be in the DMZ: databases, file servers, internal applications.

2.4.3 Zones of Risk & the Zero Trust Model

Understanding trust levels across network zones and modern security strategy

Zones of Risk

- Most networks have 2 to 4 zones of risk:
- Trusted Private LAN — Low risk, High trust. Internal devices and users.
- Extranet — Medium-low risk, Medium-high trust. Partner or supplier access.
- DMZ — Medium-high risk, Medium-low trust. Public-facing servers.
- Internet — High risk, Low trust. Completely untrusted external traffic.

Zero Trust Model

- Traditional networks automatically trust users and devices inside the organisation. This assumption is dangerous.
- A trusted insider can move freely across the network and access data they shouldn't.
- The Zero Trust model challenges this — it operates on the principle: never trust, always verify.
- Every user, device and connection is continuously verified regardless of whether they are inside or outside the network.
- Firewalls manage east-west traffic (between internal servers) and north-south traffic (data entering and leaving the network).

2.4.4 Managing Threats to the LAN

Countermeasures for common cybersecurity threats on the local area network

Threats & Countermeasures

- Unauthorized access to systems, applications and data
 - Implement strong authentication, access controls and role-based permissions.
- Unauthorized LAN access
 - Use 802.1X port-based authentication to control which devices can connect to the network.
- Exploits of data in transit
 - Encrypt all data in transit using TLS, VPNs and other encryption protocols.
- Network operating system vulnerabilities
 - Apply patches and updates regularly to network devices and operating systems.
- Unauthorized access by rogue users
 - Monitor network activity, enforce the Zero Trust model and use network segmentation.
- Unauthorized network probing and port scanning
 - Use intrusion detection systems (IDS) and firewall rules to detect and block scanning activity.

2.4 Summary — Network Hardening: Segmentation

Key concepts from this module

Segmentation Tools

- VLANs — Logically divide a network to isolate sensitive data and departments
- DMZ — Buffer zone between the internet and internal network for public-facing servers
- Firewalls — Control traffic between zones (east-west and north-south)
- Zero Trust — Never automatically trust any user or device; always verify
- Network Zones — LAN (high trust), Extranet, DMZ, Internet (no trust)

LAN Threat Countermeasures

- Strong authentication and access controls for system access
- 802.1X authentication to control LAN device access
- TLS and VPNs to encrypt data in transit
- Regular patching of network OS and devices
- Zero Trust model and segmentation to limit rogue user access
- IDS and firewall rules to detect port scanning and probing
- Key Principle: Segment first, verify always, trust nothing by default.